MIL-A-49079(AV)
AMENDMENT 1
22 July 1981

MILITARY SPECIFICATION

ALTIMETER SET, ELECTRONIC AN/APN-209(V)
(UNITS OF)

This amendment forms a part of Military Specification MIL-A-49079(AV), dated 5 December 1979, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title, line 2: Delete "AN/APN-209(V)" and substitute "AN/APN-209A(V)".

1.1, line 4: Delete "AN/APN-209(V)" and substitute "AN/APN-209A(V)".

line 8: Delete "RT-1115( )/APN-209(V)" and substitute "RT-1115A/APN-209(V)".

PAGE 3

2.1, DRAWINGS, line 2: Delete "DL-SM-B-707653" and substitute "DL-SM-B-967122".

line 3: Delete "RT-1115( )/APN-209(V)" and substitute "RT-1115A/APN-209(V)".

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3.3, line 4: Delete "DL-SM-B-707653" and substitute "DL-SM-B-967122".

line 5: Delete "RT-1115( )/APN-209(V)" and substitute "RT-1115A/APN-209(V)".

3.3.2, line 3: Delete "RT-1115( )/APN-209(V)" and substitute "RT-1115A/APN-209(V)".

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PAGE 6

3.4.10, line 7: Delete "1115( )/APN-209(V)" and substitute "1115A/APN-209(V)".

PAGE 7

3.4.11, line 5: Delete "AN/APN-209(V)" and substitute "AN/APN-209A(V)".

3.5.1, line 1: Delete "AN/APN-209(V1)" and substitute "AN/APN-209A(V)1".

3.5.2, line 1: Delete "AN/APN-209(V2)" and substitute "AN/APN-209A(V)2".

3.6.2, line 10: Under line 9 add "e. Blanking pulse".

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3.6.4, line 2: Delete "RT-1115( )/APN-209" and substitute "RT-1115A/APN-209(V)".

PAGE 9

3.6.10: Delete entire paragraph and substitute

"3.6.10 Transmitter spurious and harmonic emission method RE03 - The spurious and harmonic radiated emission from the alitmeter transmitting antenna shall be 70 dB below the level of the transmitter fundamental radiated emission (dBm/m^2) except as listed below:

4.085 - 4.515 GHz	No Limit
3.900 - 4.084 GHz	-20 dB
4.516 - 4.700 GHz	-20 dB
3.300 - 3.899 GHz	-40 dB
4.701 - 5.300 GHz	-40 dB
1.75 - 2.40 GHz	-55 dB

The spurious and harmonic radiated emission from the receiving antenna shall be less than -60 dBm except in the 4.29 GHZ to 4.31 GHz frequency range. Emission in the 4.29 GHz to 4.31 GHz frequency range shall be no greater than -17 dBm.".

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3.7.4, line 3: Delete "2000" and substitute "1000".

PAGE 11

3.7.5: Delete entire paragraph and substitute

"3.7.5 System sensitivity (see 4.9.4) - Minimum loop gain (transmit port to receive port) shall be 119 dB under standard conditions (0°C to +40°C). At temperatures other than standard conditions the minimum loop gain shall be 117 dB. The transmitter peak power shall not exceed +38.5 dBm when measured at the transmitter port.".

3.7.6, line 3: Delete "at least 3" and substitute "4 ± 0.5".

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3.7.12, line 3: Delete "1" and substitute "1.8".

3.7.13, line 4: Delete "2000" and substitute "1000".

line 6: Delete "0.55" and substitute "0.44".

3.7.16: Add this paragraph

"3.7.16 Zero altitude power (see 4.9.27) - The Receiver-Transmitter shall have a peak output power level of +27 dBm at an altitude of zero feet.".

3.8.1, line 6: Delete "-13 + 2 VDC" and substitute "-13 ± 2 VDC".

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3.8.6, line 10: Delete "300" and substitute "150".

line 11: Below line 10 add "d. Amplitude: 4.5 ± 1.0 volts peak".

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3.8.7, line 7: Delete "20" and substitute "15".

line 8: Delete "100" and substitute "75".

line 10: Delete "0.27 ± 0.05" and substitute "0.20 ± 0.06".

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line 11: Delete "1.9 ± .5" and substitue "2.7 ± 0.8".

line 12: Below line 11 add "f. Amplitude: 4.5 ± 1.0 volts peak".

3.8.8: Add this paragraph

"3.8.8 Blanking pulse (see 4.9.26) - The Receiver-Transmitter, Height Indicator shall provide a blanking pulse during the time that the transmitted RF pulse from J4 is being transmitted. The blanking pulse shall have the following characteristics:

a.	Width	2.6 ± 0.3 microseconds
b.	Amplitude	6.0 ± 0.75 volts
c.	PRF	18 + 4, -6 KHz
d.	Source Impedance On (+6V)	15 ohms maximum to +6V
e.	Source Impedance Off (0V)	1050 ohms to ground maximum
f.	Lead Time	2.0 ± 0.2 microseconds".

3.9.1, line 1: Delete "RT-1115( )/APN-209(V)" and substitute "RT-1115A/APN-209(V)".

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3.9.2, line 4: Delete "RT-1115( )/APN-209(V)" and substitute "RT-1115A/APN-209(V)".

3.10.1: Delete entire paragraph and substitute

"3.10.1 Impedance - (For reference only) The nominal antenna impedance shall be 50 ohms.".

3.10.3, line 2: Delete "10.5" and substitute "10".

3.10.4, line 2: Delete "1.2" and substitute "1.25".

line 3: Delete "1.3" and substitute "1.35".

PAGE 18

3.10.7: Delete entire paragraph.

3.10.8, line 1: Before first sentence add "(For reference only)".

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3.10.10, line 5: Delete "36231" and substitute "36440".

3.10.11, line 4: Delete "36231" and substitute "36440".

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3.13, line 4: Delete "operation" and substitute "Operational".

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TABLE I, Assemblies, line 2: Delete "RT-1115( )/APN-209(V)" and substitute "RT-1115A/APN-209(V)".

line 4: Delete "RT-1115( )/APN-209(V)" and substitute "RT-1115A/APN-209(V)".

Electrical, line 2: Delete "RT-1115( )/APN-209(V)" and substitute "RT-1115A/APN-209(V)".

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TABLE II, Electrical, line 2: Delete "RT-1115( )/APN-209(V) and substitute "RT-1115A/APN-209(V)".

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TABLE III, Subgroup I, line 1: Delete "RT-1115( )/APN-209(V)" and substitute "RT-1115A/APN-209(V)".

TABLE III, Subgroup I, line 22: After line 21 add under column headings

Inspection	Reqt Para	Insp Para
"Zero altitude power	3.7.16	4.9.27"
"Blanking pulse	3.8.8	4.9.26".

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TABLE III, Subgroup IV, line 5: Delete entire line, "Impedance, 3.10.1 and 4.17.1".

line 10: Delete entire line, "Power input, 3.10.7 and 4.17.7".

line 11: Delete entire line, "Time delay of RF signal, 3.10.8 and 4.17.8".

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4.6.1.2, line 1: Delete "RT-1115/APN-" and substitute "RT-1115A/APN-".

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4.8, line 6: Delete "RT-1115/APN-209" and substitute "RT-1115A/APN-209".

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4.8.4: Delete second sentence and substitute

"The time schedules shall be in accordance with Table 514.2-III.".

PAGE 37

4.8.11, line 6: Delete "not" and substitute "be connected to a mating connector-cable assembly or".

PAGE 38

4.9.1.2, line 23: Delete "1 V" and substitute "1.0 V".

PAGE 39

4.9.1.2, J1 pin 11 (Simulated Video), line 5: Delete "1 V" and substitute "1.0 V".

line 9: Delete "6.0" and substitute "18.0".

line 10: Delete "400" and substitute "850".

line 14: Delete "±2 nanoseconds." and substitute "±2 nanoseconds (as measured from the leading edges).".

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4.9.2.1, line 1: Delete "RT-1115( )/APN-209(V)" and substitute "RT-1115A/APN-209(V)".

4.9.3.1, line 1: Delete "RT-1115( )/APN-209(V)" and substitute "RT-1115A/APN-209(V)".

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TABLE VI, line 1: Delete "RT-1115( )/APN-209(V)" and substitute "RT-1115A/APN-209(V)".

TABLE VII, line 5: Delete "MVDC" and substitute "VDC".

line 8: Delete "tp" and substitute "to".

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4.9.4, Step 1, line 3: After "track)" add ",73 to 85 dB at temperature extremes".

Step 3, line 2: after "dB" add ",117 dB minimum at temperature extremes".

4.9.5.1, line 1: Delete "RT-1115( )/APN-209(V)" and substitute "RT-1115A/APN-209(V)".

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4.9.8.1, line 1: Delete "RT-1115( )/APN-209(V)" and substitute "RT-1115A/APN-209(V)".

4.9.9.1, line 1: Delete "RT-1115( )/APN-209(V)" and substitute "RT-1115A/APN-209(V)".

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TABLE VIII, line 1: Delete "RT-1115( )/APN-209(V)" and substitute "RT-1115A/APN-209(V)".

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4.9.13, line 2: Delete "50" and substitute "0".

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4.9.14, line 7: After line 6 add "r. J1 pin 6 blanking pulse".

4.9.15.1, line 1: Delete "RT-1115( )/APN-209(V)" and substitute "RT-1115A/APN-209(V)".

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4.9.16, line 6: Delete "2000" and substitute "1000".

4.9.17, line 7: Delete "3" and substitute "4 ± 0.5".

line 8: Delete "minimum".

4.9.18, line 6: Delete "1.0" and substitute "1.8".

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4.9.20, line 3: Delete "100" and substitute "110".

4.9.22, line 7: Delete "2000" and substitute "1000".

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4.9.26: Add this paragraph

"4.9.26 Blanking pulse (see 3.8.8). Energize the RT and verify that the blanking pulse has an amplitude of 6.0 ± 0.75 volts, a pulse width of 2.6 ± 0.3 microseconds, a PRF of 18 + 4, -6 KHz, and that the blanking pulse leads the transmitted RF pulse on J4 by 2.0 ± 0.2 microseconds when the blanking pulse is loaded down with a 10,000 ohm ± 5% resistor.".

4.9.27: Add this paragraph

"4.9.27 Zero altitude power (see 3.7.16). This test is to be performed prior to final assembly of the Receiver-Transmitter assembly and the integral indicator. Energize the Receiver-Transmitter assembly in the forced track mode and manually slew the unit for zero feet altitude. Verify that the maximum peak output power is 27 dBm or less.".

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4.13, line 1: Delete "RT-1115( )/" and substitute "RT-1115A/".

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4.17.1: Delete entire paragraph.

4.17.3, line 8: Delete entire line and substitute "antenna C as the receive antenna and B as the transmit antenna. Repeat with antenna C as the".

line 9: Delete "transmit antenna." and substitute "transmit antenna and antenna A as the receive antenna.".

line 12: Delete "g_A" and "L_{AO}" and substitute "g_B" and "L_{BO}".

line 13: Delete "g_B" and "$L_{BO}-L_{Cr}$" and substitute "g_A" and "$L_{CO}-L_{Ar}$".

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4.17.7: Delete entire paragraph.

4.17.8: Delete entire paragraph.

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4.20.1, line 1: Delete "RT-1115( )/APN-209(V)" and subsitute "RT-1115A/APN-209(V)".

PAGE 56

TABLE XI, Range column, line 7: Delete "750" above line reading "Push to test" and substitute "0".

line 11: Delete "750" above line reading "Push to test" and substitute "0".

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5.1.1.4.1, line 3: Delete "RT-1115( )/APN-209(V)" and substitute "RT-1115A/APN-209(V)".

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5.2.1, line 1: Delete "AN/APN-209" and substitute "AN/APN-209A(V)".

line 2: Delete "( )".

line 4: Delete "RT-1115( )/APN-209(V)" and substitute "RT-1115A/APN-209(V)".

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5.2.2, line 2: Delete "AN/APN-209(V)" and substitute "AN/APN-209A(V)".

line 3: DELETE "ID-1917( )APN-209(V)" and substitute "ID-1917( )/APN-209(V)".

line 4: Delete "RT-1115( )/APN-209(V)" and substitute "RT-1115A/APN-209(V)".

line 15: Delete "weitht" and substitute "weight".

line 18: Delete "P71" and substitute "P-71".

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5.2.2.1, line 7: Delete "RT-1115( )/APN-209(V)" and substitute "RT-1115A/APN-209(V)".

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6.1, line 2: Delete "209( )(V)" and substitute "209A(V)".

line 6: Delete "indentified" and substitute "identified".

line 7: Delete "APN-209(V)1" and substitute "APN-209A(V)1".

line 7: Delete "APN-209(V)2" and substitute "APN-209A(V)2".

6.2(c), line 2: Delete "4.15" and substitute "4.16".

6.2(j), line 1: Delete "(see 3.7.14)".

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6.3, line 2: Delete "4.12" and substitute "of 4.13".

line 6: Delete "4.12" and substitute "4.13".

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FIGURE 1, Note in lower left corner: Delete "RECEPTACLE SHALL BE HERMETIC SEAL TYPE".

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DD FORM 1426, DOCUMENT IDENTIFIER AND TITLE: Delete "AN/APN-209(V)" and substitute "AN/APN-209A(V)".

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